

Background

Checked tones in Southern Min are a living fossil of ancient Chinese, preserving the unreleased stops that Mandarin lost.

Table 1. Invitation to Wine 將進酒 (Li Bai, Tang) — Southern Min preserves the Tang rhyme; Mandarin breaks it.

Line	Translation	Southern Min	Mandarin
陳王昔時宴平樂	The prince once feasted at Pingle	-lok	lè
斗酒十千恣歡謔	Ten thousand coins for wine and revelry	-iok	xuè
主人何為言少錢	Why does the host speak of having little?	— (no rhyme)	— (no rhyme)
徑須沽取對君酌	Just buy the wine and drink with me	-tiok	zhuó

Research Questions

Southern Min has two checked tones — Tone 3 (Low / Falling) and Tone 5 (High / Flat), ending in four unreleased stops: /k/, /p/, /t/, /ʔ/. They are acoustically subtle: airflow is blocked but never audibly released, leaving only sub-segmental cues.

PERCEPTION

How do listeners across L1 backgrounds perceive these contrasts?

PRODUCTION

How do Southern Min checked-tone codas contrast acoustically?

Conclusion

Acoustically, checked-tone codas are significantly distinct from each other, relying on multiple cues. /ʔ/ is the most acoustically distinct, while /k/ is the least distinct. Perceptually, listeners hear what their L1 has trained them to hear; however, native speakers underperform, which is unusual and counterintuitive, yet explainable.

Perception

CROSS-LINGUISTIC LISTENER STUDY

Listener pool at a glance

65 participants across 5 L1 groups

40 forced-choice trials (16 tone + 24 coda)

5 L1s tested: Mandarin 13 / Korean 9 / English 14 / Japanese 13 / S. Min 16

Figure 1 · Group accuracy by listener L1

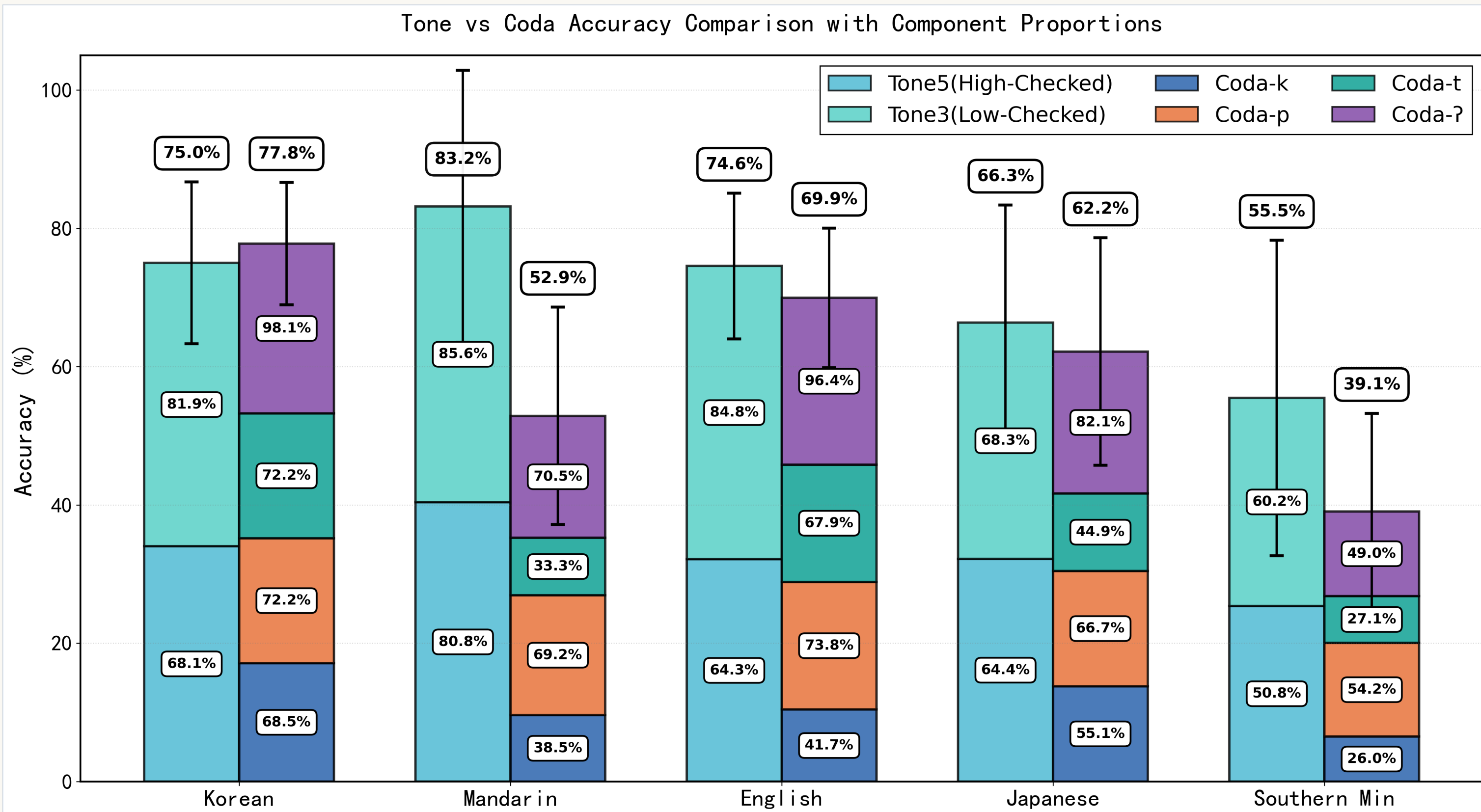


Figure 1. Mean tone and coda identification accuracy across language groups with error bars.

Group Rankings (> means $p < .05$; $\approx$ means not significant (N.S.))

TONE Mandarin > Korean $\approx$ English > Japanese > S. Min | CODA Korean $\approx$ English > Japanese > Mandarin > S. Min

Table 2 · Eight perception findings

Finding	Pattern	Note
① Mandarin tonal advantage	Highest tone-identification	L1 tonal knowledge transfers to checked-tone identification.
② Korean & English coda edge	Highest coda-identification	F2/F3 vowel-offset sensitivity drives the coda advantage.
③ Tone 3 > Tone 5	All 5 groups	Low-checked tone consistently easier to identify.
④ /ʔ/ universally salient	High accuracy	Glottal stop perceptually distinct.
⑤ /k/, /p/, /t/ confused	Low accuracy	Three oral places mutually confused across all groups.
⑥ F2 → /t/ identification	Korean & English advantage	Higher F2 sensitivity → better /t/ accuracy.
⑦ F3 → /k/ identification	Korean & Japanese advantage	F3-region sensitivity → better /k/ accuracy.
⑧ Southern Min Paradox	Tone 55.5% · Coda 39.1%	Native speakers underperform. See explanation below.

Why do natives underperform? Three possible explanations

- ① Variable realizations of checked-tone codas in modern Southern Min: stop closure, energy dip, irregular glottal vibration, and coda deletion.
- ② Free variation in coda place — e.g., 合 (hap / hak), 達 (tat / tap), 結 (kat / kak).
- ③ Lexical access dominates: disyllabic context reduces reliance on isolated coda cues.

Production

ACOUSTIC ANALYSIS · 4 SPEAKERS · 96 TOKENS

Figure 2-3 · Formant trajectories by coda & Duration by coda

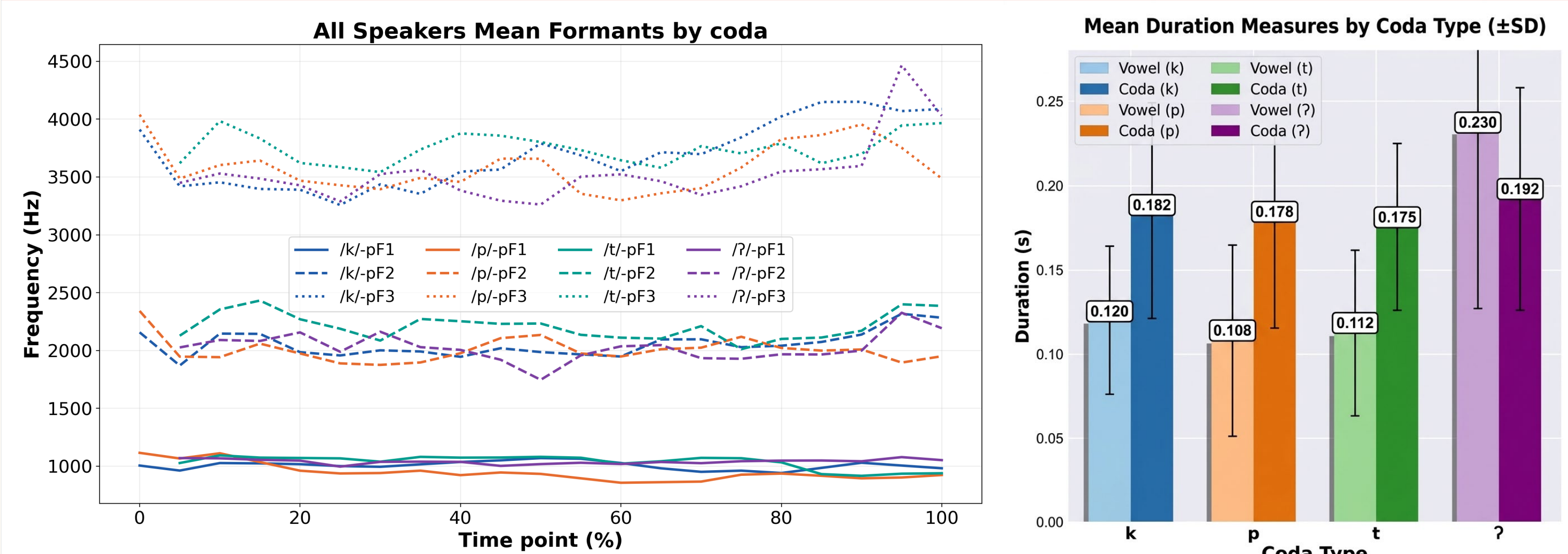


Figure 2. F1–F3 trajectories across normalized time, by coda.

Figure 3. Vowel and coda duration by coda.

Figures 4–7 · Pitch, Energy, Spectral tilt & CPP

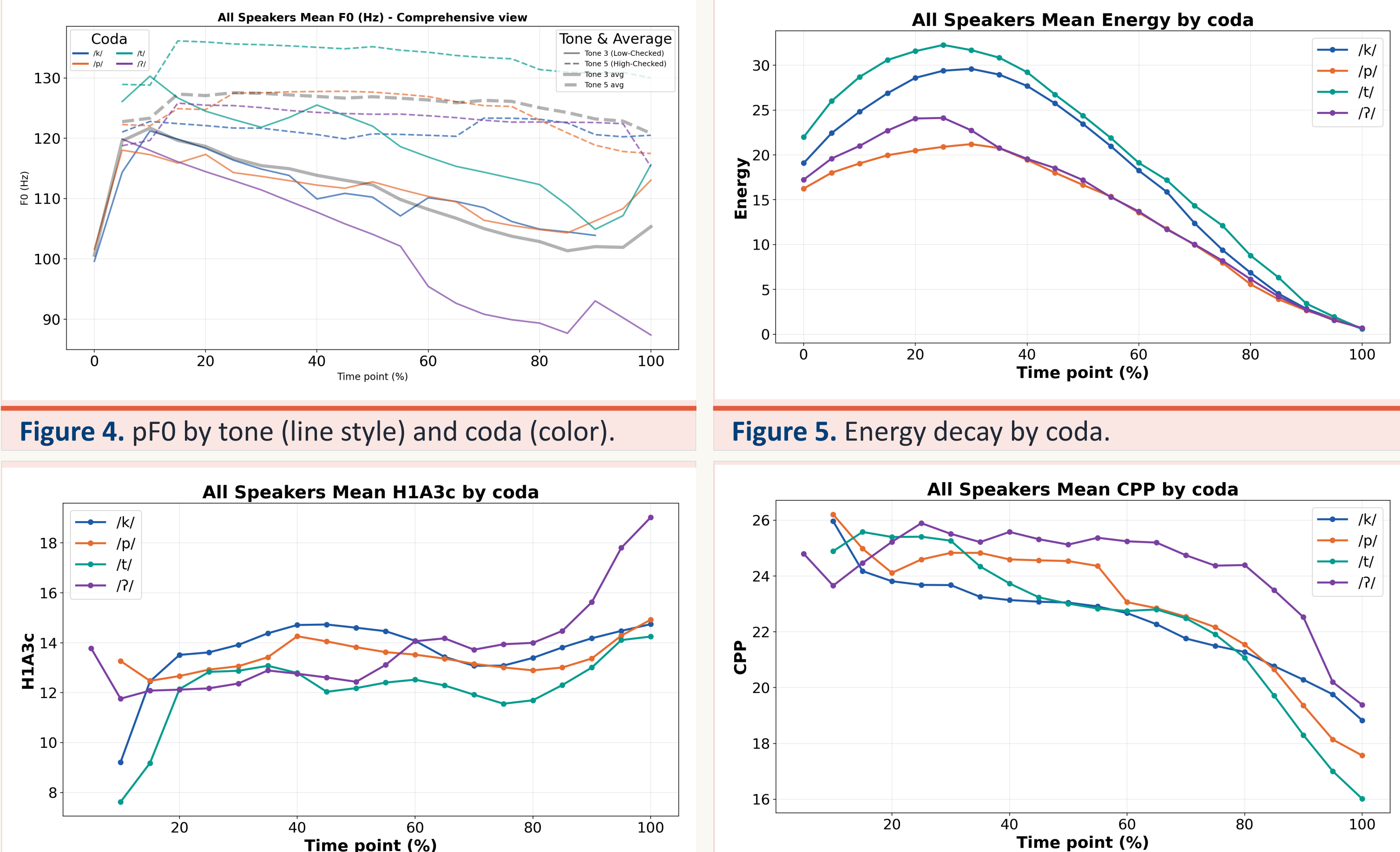


Figure 4. pF0 by tone (line style) and coda (color).

Figure 5. Energy decay by coda.

Figure 6. H1*–A3* spectral tilt by coda.

Figure 7. Cepstral Peak Prominence (CPP) by coda.

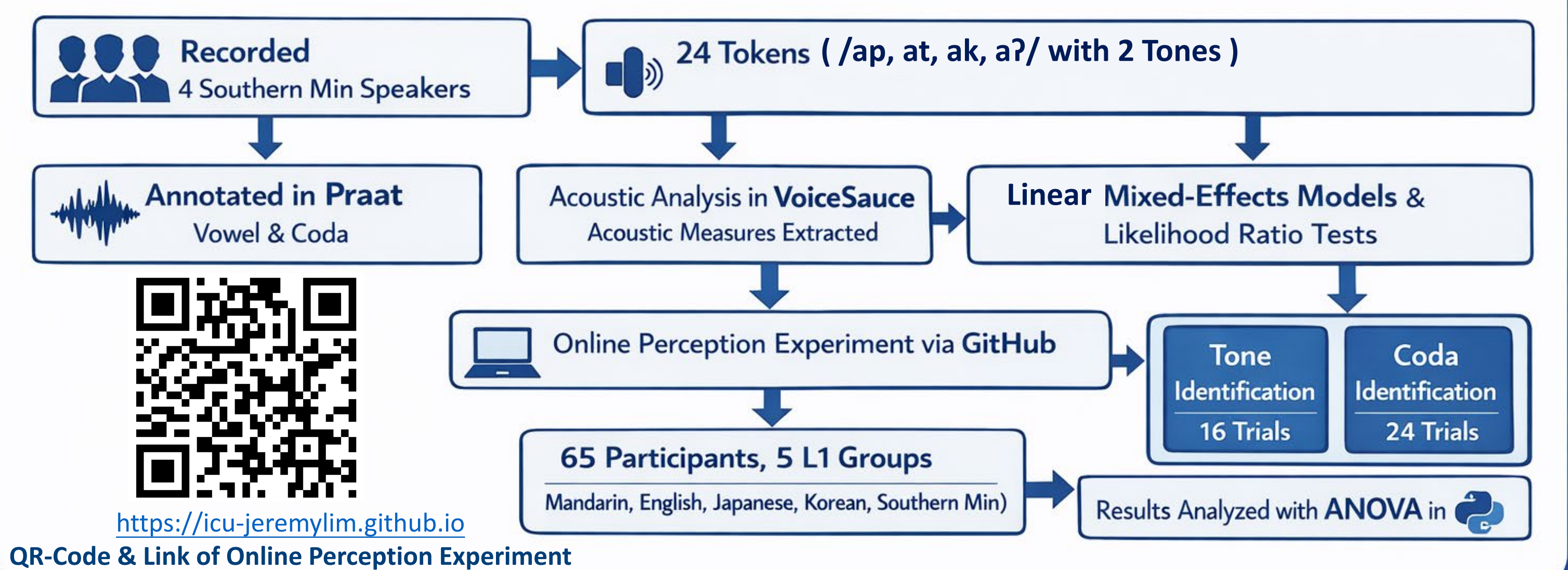
Table 3 · Acoustic measure summary

Parameter	Time Window	Key finding
F0	40–100 %	tone 3 vs. 5 $p < .001$; coda pairs $p < .05$.
F1	40–60 %	/p/ lower than other codas $p < .01$.
F2	40–60 %	/t/ higher than other codas $p < .05$.
F3	60–100 %	/k/ higher than other codas $p < .05$.
Energy	20–80 %	coda contrast $p < .05$; /p/ & /ʔ/ faster decay.
CPP	40–80 %	tone $p < .001$; coda pairs $p < .05$.
H1*–A3*	60–100 %	/t/ lower in tone 5 $p < .05$.
Duration	Vowel	/ʔ/ longest vowel $p < .001$.

Take Home Message

Every language is a treasure. In Thai & Korean, natives excel at both speaking and perceiving their unreleased codas. In Southern Min, however, the result is surprising: natives can produce unreleased codas accurately, but cannot perceive them as well. Similar sounds, different results, revealing different cognitive patterns through diverse human languages.

Methodology



References

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Acknowledgement

This is my first linguistic research. Thank you for reading this poster. Any advice or criticism is warmly welcomed. Your presence here is my joy, honor, and motivation to keep walking this path. I am grateful to the HISPhonCog 2026 organizers for this valuable opportunity to present. Special thanks to my kind teachers, Dr. Seunghun Lee, Dr. Hyun Kyung Hwang, and Dr. Hiroki Tanaka, who have witnessed the incubation of this work and offered insightful guidance. I also thank the 65 participants of the online perception experiments and the 4 Southern Min speakers who lent their voices for recording, as well as Claude and ChatGPT for assistance with brainstorming, coding, and poster design. To my family and friends, thank you for supporting me in countless ways throughout this journey. Above all, glory to Jesus Christ, my Lord and Savior, the Bread of Life. For His glory, faithful missionaries undertook the earliest dictionaries, romanization systems, and linguistic studies of Southern Min. Their testimonies inspire me to walk the same path — preserving understudied languages and sharing God's word in every tongue that does not yet have a Bible.

Paper & Data



<https://github.com/ICU-JeremyLim/HISPhonCog2026Paper141>